

HDOC DAY-25 ASSIGNMENT

Menu-Driven Decreasing Patterns Using Switch-Case

1. Problem Understanding

The task is to write a menu-driven C program using switch-case to display four decreasing patterns. The program uses nested loops without arrays. Each pattern is implemented in a separate parameterless user-defined function with void return type. The selected function reads the number of rows, n, and prints the required pattern.

- Decreasing right-angled triangle using asterisks.
- Decreasing natural-number pattern.
- Decreasing repeated-number pattern.
- Reverse-number triangle.

2. Standard Test Cases

Test Case	Choice	Input (n)	Expected Output	Status
TC1	1	4	**** / **** / ** / *	Pass
TC2	1	1	*	Pass
TC3	2	4	1 2 3 4 / 1 2 3 / 1 2 / 1	Pass
TC4	2	2	1 2 / 1	Pass
TC5	3	4	4 4 4 4 / 3 3 3 / 2 2 / 1	Pass
TC6	3	2	2 2 / 1	Pass
TC7	4	4	4 3 2 1 / 3 2 1 / 2 1 / 1	Pass
TC8	4	2	2 1 / 1	Pass

3. Algorithm

A. Common Menu Logic

1. Start.
2. Display the four menu options.
3. Read the user's choice.
4. Use switch-case to select the required pattern.
5. Call the corresponding void function.
6. The selected function reads n and prints the pattern using nested loops.
7. If the choice is invalid, display "Invalid choice".
8. Stop.

B. Decreasing Right-Angled Triangle

1. Start.
2. Read the number of rows n.
3. Use an outer loop with i starting from n and decreasing to 1.
4. For each row, use an inner loop from 1 to i.
5. Print "*" in the inner loop.
6. After each row, print a newline.
7. Stop.

C. Decreasing Natural-Number Pattern

1. Start.
2. Read the number of rows n.
3. Use an outer loop with i starting from n and decreasing to 1.
4. For each row, use an inner loop with j from 1 to i.
5. Print j followed by a space.
6. After each row, print a newline.
7. Stop.

D. Decreasing Repeated-Number Pattern

1. Start.
2. Read the number of rows n.
3. Use an outer loop with i starting from n and decreasing to 1.
4. For each row, use an inner loop from 1 to i.
5. Print the value of i followed by a space.
6. After each row, print a newline.
7. Stop.

E. Reverse-Number Triangle

1. Start.
2. Read the number of rows n.
3. Use an outer loop with i starting from n and decreasing to 1.
4. For each row, use an inner loop with j starting from i and decreasing to 1.
5. Print j followed by a space.
6. After each row, print a newline.
7. Stop.

4. C Program

```
#include <stdio.h>

void decreasingStars()
{
    int n, i, j;
    printf("Enter number of rows: ");
```

```

scanf("%d", &n);
for (i = n; i >= 1; i--)
{
    for (j = 1; j <= i; j++)
        printf("* ");
    printf("\n");
}
}

void decreasingNatural()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void decreasingRepeated()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", i);
        printf("\n");
    }
}

void reverseTriangle()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)

```

```

    {
        for (j = i; j >= 1; j--)
            printf("%d ", j);
        printf("\n");
    }
}

int main()
{
    int choice;
    printf("MENU\n");
    printf("1. Decreasing right-angled triangle\n");
    printf("2. Decreasing natural-number pattern\n");
    printf("3. Decreasing repeated-number pattern\n");
    printf("4. Reverse-number triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1: decreasingStars();
        break;
        case 2: decreasingNatural();
        break;
        case 3: decreasingRepeated();
        break;
        case 4: reverseTriangle();
        break;
        default: printf("Invalid choice\n");
    }
    return 0;
}

```

5. Compilation and Execution Evidence

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ gcc question.c -o question
(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 4
* * *
* *
*
*

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 1
*

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1 2
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: 2
1 2
1

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 4
4 4 4 4
3 3 3
2 2
1

(kali@kali)-[~/Desktop]
$ ./question
```

```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~/Desktop
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 2
2 2
1

(kali@kali)~/Desktop
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: 4
4 3 2 1
3 2 1
2 1
1

(kali@kali)~/Desktop
$ ./question
MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: 2
2 1
1

(kali@kali)~/Desktop
$

```

6. Verification of All Test Cases

Test Case	Choice	Input	Expected Result	Actual Result	Status
TC1	1	4	**** / *** / ** / *	**** / *** / ** / *	Pass
TC2	1	1	*	*	Pass
TC3	2	4	1 2 3 4 / 1 2 3 / 1 2 / 1	1 2 3 4 / 1 2 3 / 1 2 / 1	Pass
TC4	2	2	1 2 / 1	1 2 / 1	Pass
TC5	3	4	4 4 4 4 / 3 3 3 / 2 2 / 1	4 4 4 4 / 3 3 3 / 2 2 / 1	Pass
TC6	3	2	2 2 / 1	2 2 / 1	Pass
TC7	4	4	4 3 2 1 / 3 2 1 / 2 1 / 1	4 3 2 1 / 3 2 1 / 2 1 / 1	Pass
TC8	4	2	2 1 / 1	2 1 / 1	Pass
